



**INDIAN INSTITUTE OF TECHNOLOGY ROPAR**

Department of Mechanical Engineering

**Course Project Report**

**FEM (ME-515)**

Session: 2025–2026 • Semester II

**Title of the Project:**

**Finite Element Analysis of a Cantilever Beam: Static,  
Modal and Convergence Study**

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## Abstract

This project presents a comprehensive finite element analysis (FEM) of a cantilever beam subjected to a distributed transverse load, encompassing both static and modal analysis using a custom-developed 2D finite element code. The beam is modeled as a plane stress problem discretized using four-node rectangular (bilinear quadrilateral) elements, with the code designed to accept variable domain dimensions, mesh density in both the x- and y-directions, and material properties as user-defined inputs.

The mesh determined from the static convergence study is adopted as the baseline, and an additional mesh convergence analysis is performed targeting the fifth natural frequency to ensure modal accuracy. Natural frequencies are compared with analytical predictions from Euler–Bernoulli beam theory, and the evolution of mode shape complexity with increasing mode number is discussed.

The study demonstrates the capability and accuracy of the developed FEM code in capturing both the static deformation behavior and the free vibration characteristics of cantilever beams, and highlights the importance of appropriate mesh refinement and beam theory selection as a function of beam slenderness.

## 1. Static Analysis

### 1.1 Governing Equations and Domain Assumptions

A cantilever beam of dimensions  $a \times h$  is considered. The out-of-plane thickness is very small, so the problem is modeled under plane stress. The constitutive relation (Hooke's law) is:

$$\sigma = D \varepsilon$$

where  $\sigma = [\sigma_{xx}, \sigma_{yy}, \tau_{xy}]^T$  and  $\varepsilon = [\varepsilon_{xx}, \varepsilon_{yy}, \gamma_{xy}]^T$ . The plane-stress constitutive matrix is:

$$D = E / (1 - \nu^2) \times \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1 - \nu)/2 \end{bmatrix}$$

### 1.2 Isoparametric Formulation (4-Node Quadrilateral Element)

The domain is discretized using 4-node bilinear quadrilateral (Q4) elements. The mapping from natural coordinates  $(\xi, \eta)$  to physical coordinates  $(x, y)$  and the displacement interpolation both use the same shape functions:

$$\begin{aligned} N_1 &= \frac{1}{4} (1 - \xi) (1 - \eta), & N_2 &= \frac{1}{4} (1 + \xi) (1 - \eta), & N_3 &= \frac{1}{4} (1 + \xi) (1 + \eta), \\ N_4 &= \frac{1}{4} (1 - \xi) (1 + \eta) \end{aligned}$$

### 1.3 Strain–Displacement Matrix and Jacobian

Physical-space derivatives are obtained via the Jacobian  $J$ . The strain–displacement matrix  $B$  relates nodal displacements  $d^e$  to strains  $(\varepsilon = B d^e)$ . For node  $i$ :

$$B_i = [\partial N_i / \partial x, 0; 0, \partial N_i / \partial y; \partial N_i / \partial y, \partial N_i / \partial x] \rightarrow B = [B_1 \ B_2 \ B_3 \ B_4]$$

### 1.4 Element Stiffness Matrix (Gauss Quadrature)

The element stiffness matrix is integrated numerically using  $2 \times 2$  Gauss quadrature ( $W_\square = 1$ , points at  $\pm 1/\sqrt{3}$ ):

$$K^e = \iint B^T D B \det(J) d\xi d\eta \approx \sum_{\square} B^T(\xi_{\square}, \eta_{\square}) D B(\xi_{\square}, \eta_{\square}) t \det(J) W_{\square}$$

### 1.5 Load Vector for Distributed Load

A distributed load  $q$  acts on the top edge ( $\eta = 1$ ). The consistent load vector per element is:

$$f^e = \int N^T(\xi, 1) [0, -q_a]^T (L^e/2) d\xi \quad \text{where} \quad q_a = q/t$$

### 1.6 Global Assembly and Boundary Conditions

The global system  $KU = F$  is assembled from element contributions. For the cantilever, all DOFs at  $x = 0$  are fixed:

$$u(0, y) = 0, \quad v(0, y) = 0$$

## 1.7 Analytical Validation

### (a) Euler–Bernoulli Beam Theory

Transverse deflection (neglects shear; valid for slender beams,  $AR \geq 10$ ):

$$v(x) = qx^2(x^2 + 6a^2 - 4ax) / (24EI)$$

### (b) Timoshenko Beam Theory

Adds a shear correction term (important for thick beams,  $AR < 5$ ):

$$v(x) = v^{\text{Luler}}(x) + qx(2a-x) / (2\kappa GA) \quad \text{where} \quad G = E / [2(1+\nu)]$$

## 2. Mesh Convergence Analysis & Comparison with Beam Theories

### 2.1 Objective

Mesh convergence analysis was performed to ensure the finite element solution becomes independent of mesh refinement. Two quantities were monitored as the number of elements was increased:

- Vertical displacement at point A ( $x = L/2$ )
- Total strain energy of the system

### 2.2 Input Parameters

| Parameter                 | Value    |
|---------------------------|----------|
| Length (L)                | 1 m      |
| Height (H)                | 0.1 m    |
| Thickness (t)             | 0.01 m   |
| Young's Modulus (E)       | 210 GPa  |
| Poisson's Ratio ( $\nu$ ) | 0.28     |
| Distributed Load (q)      | 5000 N/m |

### 2.3 Nodal Displacements

The transverse displacement ( $U_y$ ) remains approximately uniform across the thickness at a given cross-section — the beam section undergoes predominantly rigid-body translation in the vertical direction. The longitudinal displacement ( $U_x$ ) exhibits a linear variation across the thickness, passing through zero at the neutral axis. This represents tensile deformation in the upper fibres and compressive deformation in the lower fibres, consistent with classical beam theory.

## 2.4 Displacement Convergence

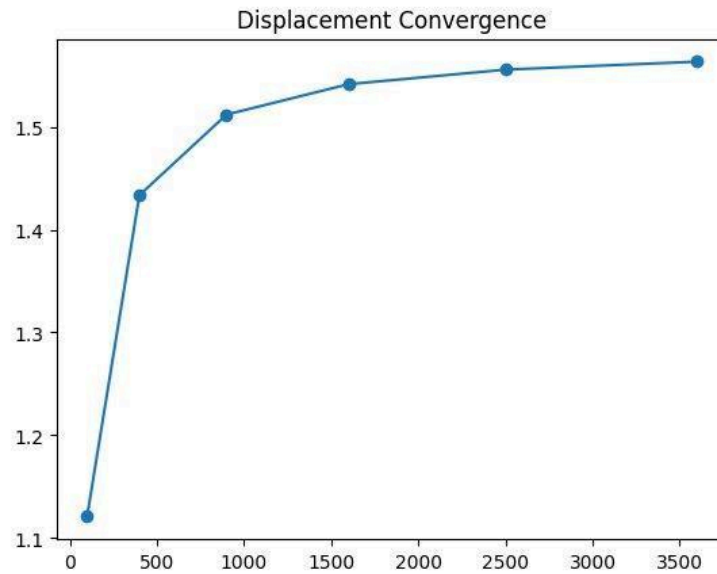


Figure 1. Displacement convergence — tip displacement (normalised) vs. number of elements

## 2.5 Mesh Convergence Results

| Elements | FEM Tip (m) | Euler (m) | Error (%) |
|----------|-------------|-----------|-----------|
| 160      | 3.495e-03   | 3.571e-03 | 2.14      |
| 360      | 3.554e-03   | 3.571e-03 | 0.49      |
| 640      | 3.575e-03   | 3.571e-03 | 0.11      |
| 1000     | 3.585e-03   | 3.571e-03 | 0.39      |

## 2.6 Aspect Ratio Study

| AR (L/H) | FEM (m)   | Euler (m) | Error (%) |
|----------|-----------|-----------|-----------|
| 2        | 7.146e-6  | 5.714e-6  | 25.06     |
| 5        | 2.309e-4  | 2.232e-4  | 3.42      |
| 10       | 3.585e-3  | 3.571e-3  | 0.39      |
| 20       | 5.696e-2  | 5.714e-2  | 0.31      |
| 50       | 2.222e+00 | 2.232e+00 | 0.48      |

## 2.7 Validation Plot

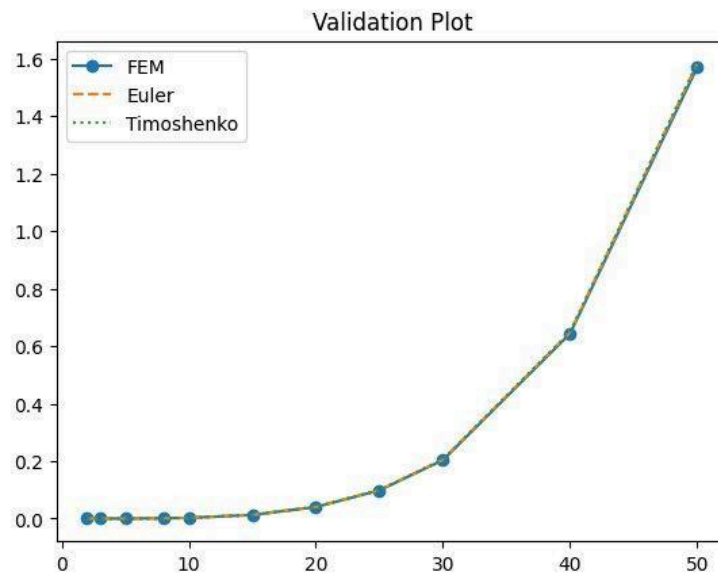


Figure 2. Validation plot — FEM vs. Euler–Bernoulli vs. Timoshenko tip displacement vs. aspect ratio

The FEM–Euler error is high at low aspect ratios due to significant shear deformation neglected by Euler theory. As the aspect ratio increases shear effects diminish and FEM converges to the Euler solution. A slight error increase at very high aspect ratios is due to mesh distortion, mitigated by keeping element aspect ratio  $dx/dy$  close to unity.

## 3. Energy Convergence

The energy convergence plot shows that total strain energy increases with mesh refinement and asymptotes to a constant value, confirming that the solution becomes mesh-independent beyond a sufficient refinement level.

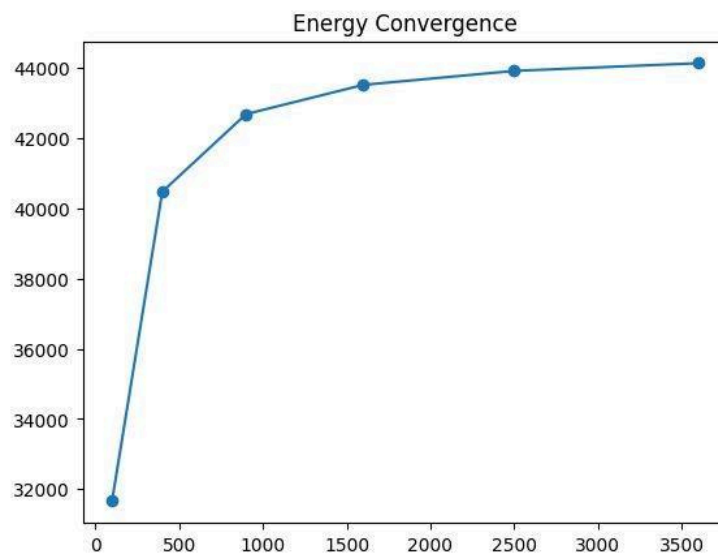


Figure 3. Energy convergence — total strain energy (J) vs. number of elements

### 3.1 Effect of Aspect Ratio on Strain Energy

Beams with higher aspect ratios store significantly more strain energy: slender beams undergo larger deflections under the same load, increasing accumulated strain. Low-aspect-ratio beams are stiffer and store less energy.

### 3.2 Physical Interpretation

Strain energy is directly proportional to structural deformation. The results clearly show the transition from stiff, shear-dominated behavior at low AR to flexible, bending-dominated behavior at high AR, validating both the code and theoretical expectations.

## 4. Stress and Strain

### 4.1 Normal Stress $\sigma_x$ (Bending Stress)

The normal stress  $\sigma_x$  exhibits a linear variation across the beam thickness — tension (+ve) in the top fibres and compression (−ve) in the bottom fibres. The maximum magnitude occurs at the fixed support ( $x = 0$ ) where the bending moment is greatest, and decreases to zero at the free tip. This is fully consistent with Euler–Bernoulli bending theory.

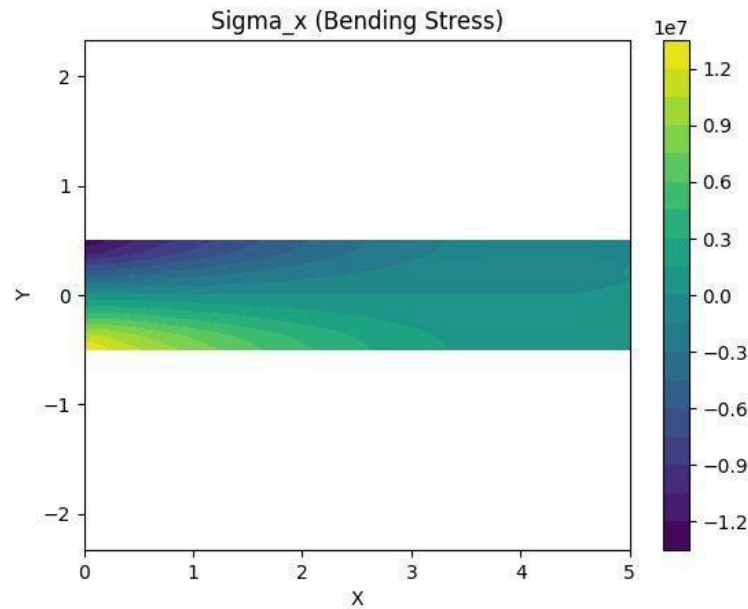


Figure 4.  $\sigma_x$  (Bending Stress) contour — linear variation across the cross-section ( $\text{Pa}, \times 10^7$ )

### 4.2 Normal Stress $\sigma_y$

Transverse normal stress  $\sigma_y$  is significantly smaller in magnitude compared to  $\sigma_x$ . It is concentrated near the fixed boundary due to constraint effects and is essentially negligible over most of the beam domain, confirming the dominance of bending stress.

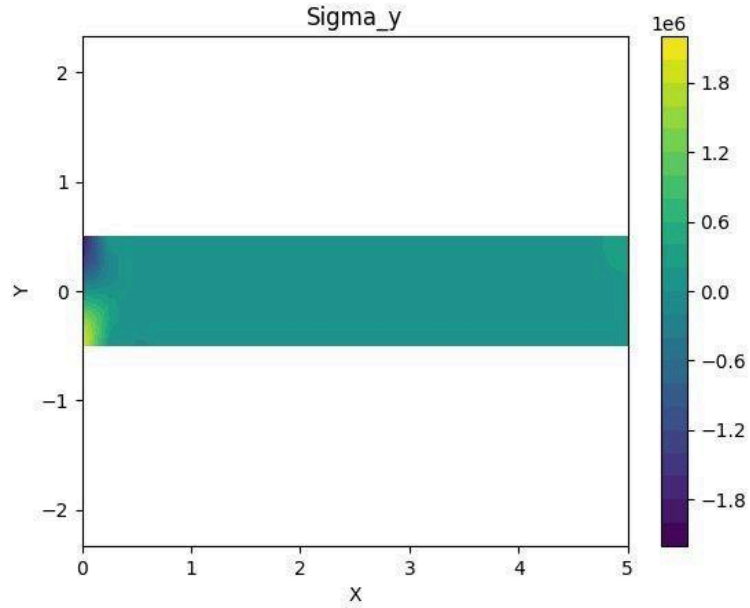


Figure 5.  $\sigma_y$  contour — small transverse stress, significant only near the clamped end (Pa,  $\times 10^6$ )

#### 4.3 Shear Stress $\tau_{xy}$

Shear stress  $\tau_{xy}$  is maximum near the fixed support and decreases progressively along the beam length. The distribution is symmetric about the neutral axis with the maximum at the beam centre, consistent with classical beam theory where shear is dominant near supports.

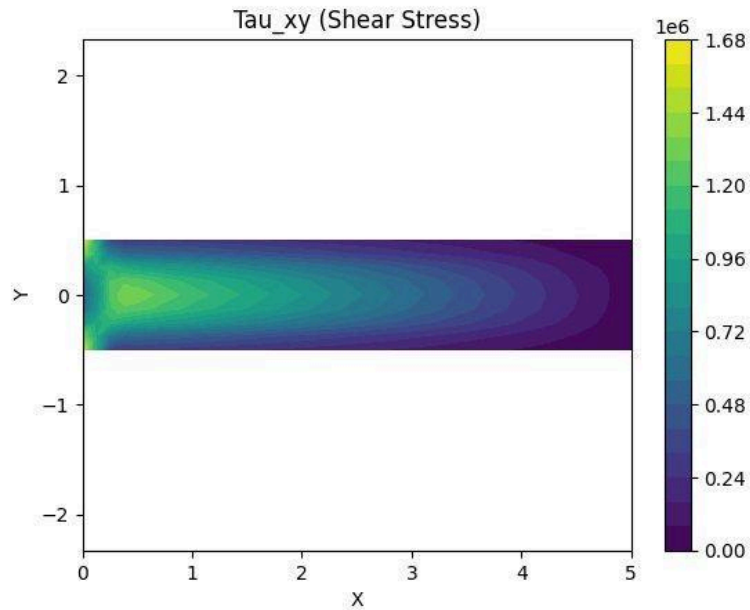


Figure 6.  $\tau_{xy}$  (Shear Stress) contour — maximum near the clamped end (Pa,  $\times 10^6$ )

#### 4.4 Shear Stress Distribution at Mid-section

The cross-sectional shear stress distribution at mid-span ( $x = L/2$ ) shows a near-parabolic profile — maximum at the neutral axis and reducing toward the outer surfaces. Minor deviations from a perfect parabola arise from the use of bilinear (4-node) elements and nodal stress averaging.

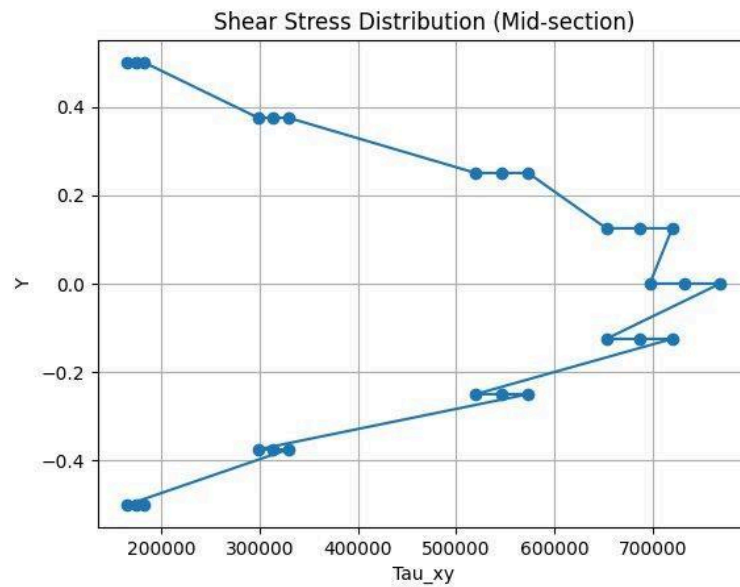


Figure 7. Shear stress distribution across the cross-section at mid-span

#### 4.5 Normal Strain $\epsilon_x$

The normal strain  $\epsilon_x$  mirrors the bending stress pattern: linear variation across the thickness (tension at top, compression at bottom, zero at the neutral axis), with maximum values at the fixed support decreasing toward the free end.

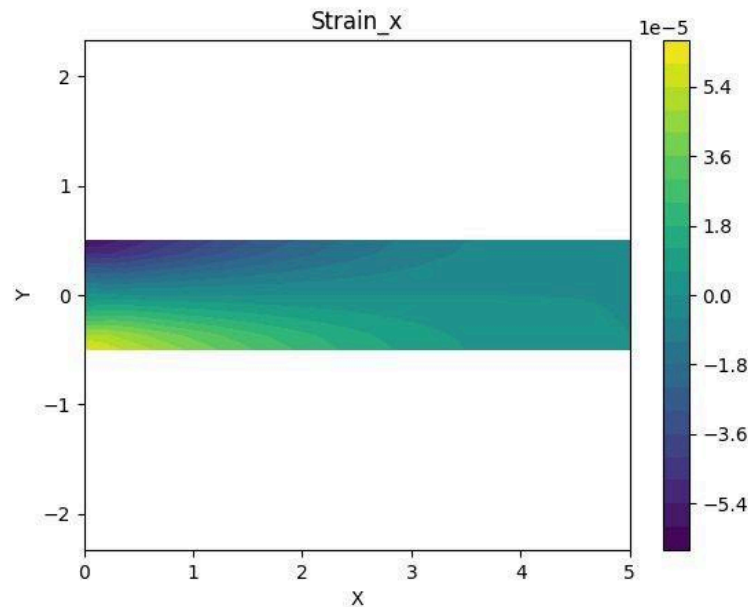


Figure 8.  $\epsilon_x$  (Normal Strain in x) contour — linear bending strain distribution ( $\times 10^{-5}$ )

#### 4.6 Transverse Strain $\epsilon_y$

The transverse strain  $\epsilon_y$  is significantly smaller than  $\epsilon_x$ . Its relation to the bending strain through  $\epsilon_y = -\nu\epsilon_x$  (Poisson coupling) is visible in the contour: where  $\epsilon_x$  is tensile at the top,  $\epsilon_y$  is compressive, and vice-versa. This validates the plane stress assumption.

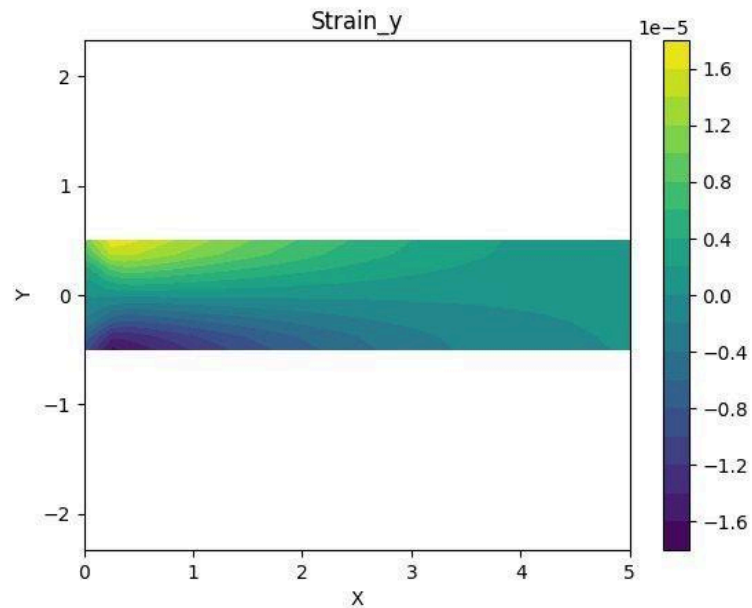


Figure 9.  $\epsilon_y$  (Normal Strain in y) contour — Poisson effect clearly visible ( $\times 10^{-5}$ )

#### 4.7 Shear Strain $\gamma_{xy}$

The shear strain  $\gamma_{xy}$  is maximum near the fixed support and at the neutral axis, consistent with the shear stress distribution. It diminishes along the beam length and toward the outer fibres. The contour pattern closely mirrors  $\tau_{xy}$  divided by the shear modulus  $G$ .

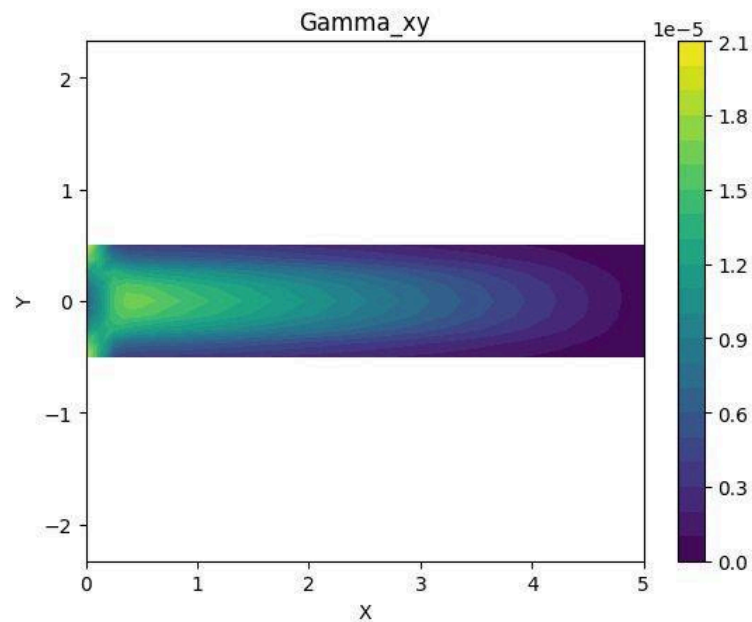


Figure 10.  $\gamma_{xy}$  (Shear Strain) contour — maximum near clamped end and neutral axis ( $\times 10^{-5}$ )

#### 4.8 Von Mises Stress

The Von Mises stress combines all stress components into a single scalar failure criterion. It is highest near the fixed support at the top and bottom fibres (where bending stress is maximum) and decreases toward the free tip. This distribution correctly identifies the fixed root as the critical failure location, consistent with the maximum bending moment location.

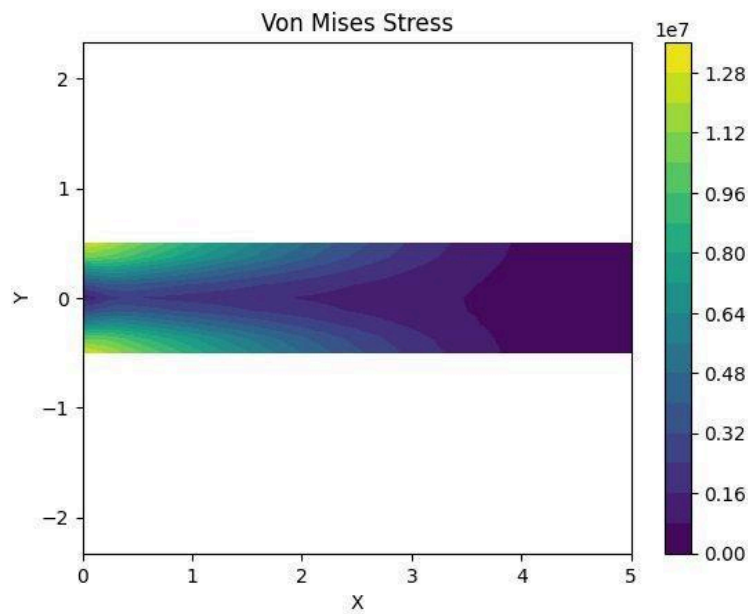


Figure 11. Von Mises stress contour — critical stress at the clamped root (Pa,  $\times 10^7$ )

## 5. Tip Displacement Validation

The maximum displacement at the free end (beam tip) serves as the primary validation parameter against the analytical solution.

| Source                       | Tip Displacement (m)   |
|------------------------------|------------------------|
| Analytical (Euler–Bernoulli) | $3.571 \times 10^{-3}$ |
| FEM Code (640 elements)      | $3.575 \times 10^{-3}$ |
| SolidWorks Simulation        | $3.608 \times 10^{-3}$ |

The FEM result with 640 elements matches the analytical value to within 0.11%, confirming excellent accuracy. The SolidWorks result (~1% higher) is attributed to triangular mesh elements, nodal averaging in post-processing, and higher-order deformation effects. All three results agree closely, validating the developed FEM code.

## 6. Modal Analysis

### 6.1 Input Parameters

| Parameter       | Value   |
|-----------------|---------|
| Length (L)      | 1 m     |
| Height (H)      | 0.1 m   |
| Elements in x   | 100     |
| Elements in y   | 10      |
| Young's Modulus | 210 GPa |
| Poisson's Ratio | 0.28    |

|                    |                        |
|--------------------|------------------------|
| Thickness          | 0.01 m                 |
| Density ( $\rho$ ) | 7700 kg/m <sup>3</sup> |

## 6.2 Natural Frequencies

The first six natural frequencies extracted from the FEM modal analysis (eigenvalue problem  $K\phi = \omega^2 M\phi$ ) are:

| Mode | FEM Frequency (Hz) | Mode Type            | Zero Crossings |
|------|--------------------|----------------------|----------------|
| 1    | 17.99              | 1st Flexural Bending | 0              |
| 2    | 112.57             | 2nd Flexural Bending | 1              |
| 3    | 314.42             | 3rd Flexural Bending | 2              |
| 4    | 613.96             | 4th Flexural Bending | 3              |
| 5    | 1010.40            | 5th Flexural Bending | 4              |
| 6    | 1297.77            | 1st Axial Mode       | —              |

## 6.3 Mode Shapes

### Mode 1 ( $f = 17.99$ Hz) — 1st Flexural Bending

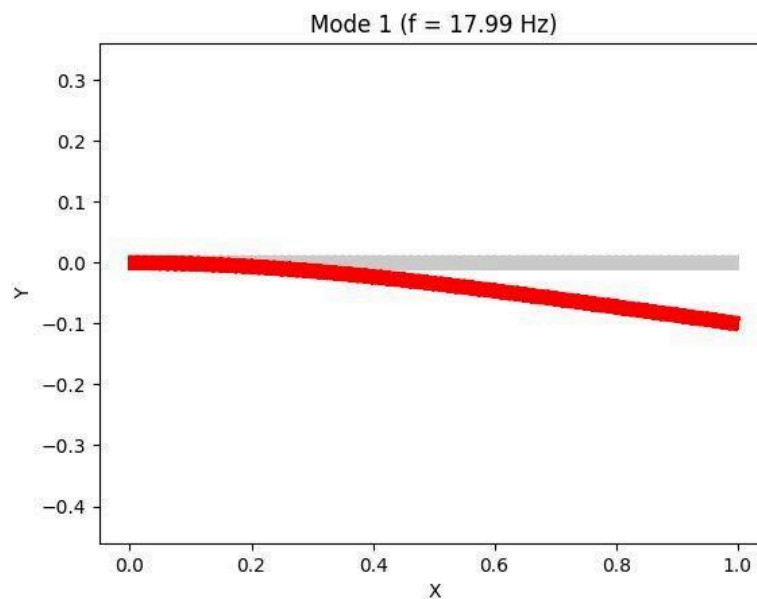


Figure 12. Mode 1: fundamental flexural bending at 17.99 Hz

Observation: A smooth, monotonically curved shape with no internal zero-crossings. Maximum transverse displacement occurs at the free end. This is the classical first bending mode of a cantilever beam.

### Mode 2 ( $f = 112.57$ Hz) — 2nd Flexural Bending

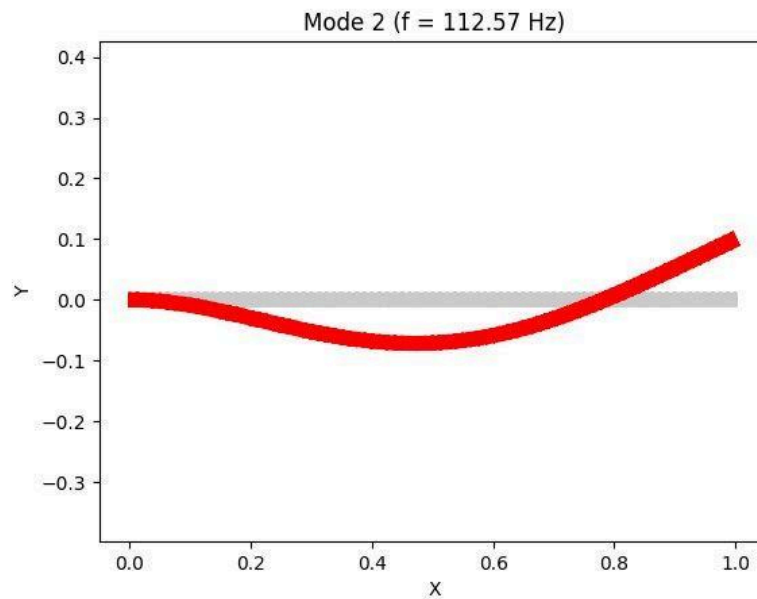


Figure 13. Mode 2: second flexural bending at 112.57 Hz

Observation: One internal zero-crossing and one curvature reversal, producing an S-curve shape. Deformation remains bending-dominated.

#### Mode 3 ( $f = 314.42$ Hz) — 3rd Flexural Bending

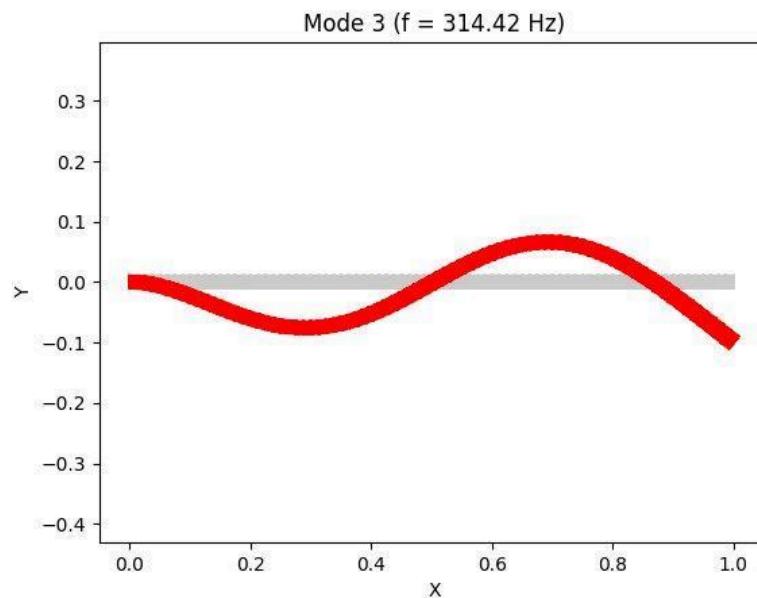


Figure 14. Mode 3: third flexural bending at 314.42 Hz

Observation: Two internal zero-crossings. The sinusoidal shape with increasing spatial frequency is clearly visible, indicating higher-harmonic bending.

#### Mode 4 ( $f = 613.96$ Hz) — 4th Flexural Bending

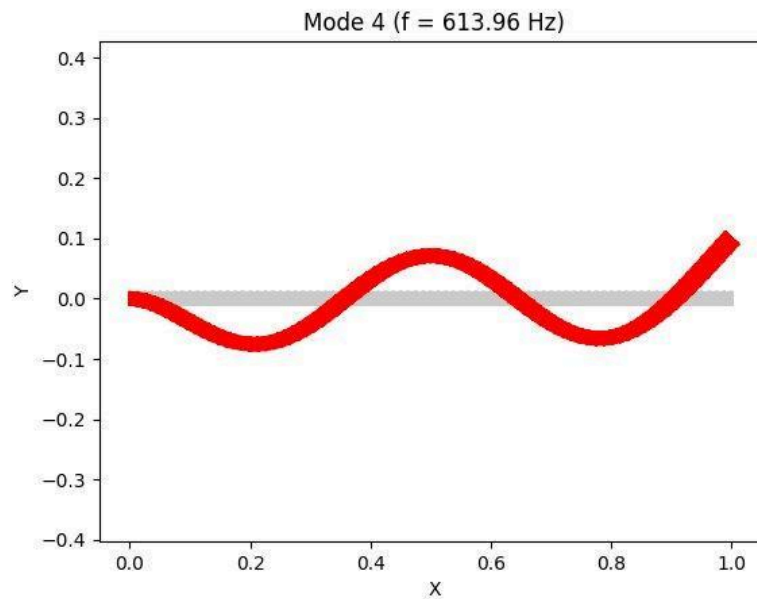


Figure 15. Mode 4: fourth flexural bending at 613.96 Hz

Observation: Three internal zero-crossings with a pronounced wave-like pattern. Deformation complexity increases significantly with the anti-nodes having comparable amplitudes.

#### Mode 5 (f = 1010.40 Hz) — 5th Flexural Bending

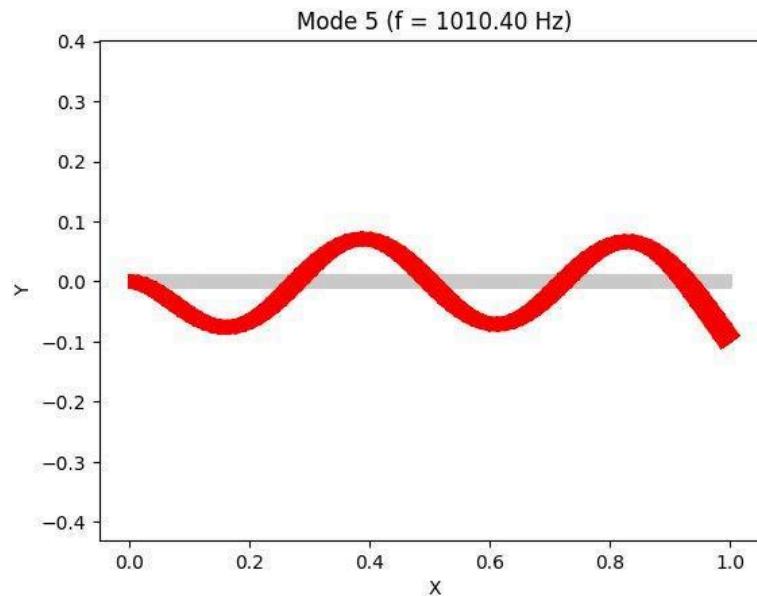


Figure 16. Mode 5: fifth flexural bending at 1010.40 Hz

Observation: Four internal zero-crossings with multiple peaks and valleys. This is a complex higher-order bending mode with short wavelengths requiring sufficient mesh density for accurate resolution.

#### Mode 6 (f = 1297.77 Hz) — 1st Axial Mode

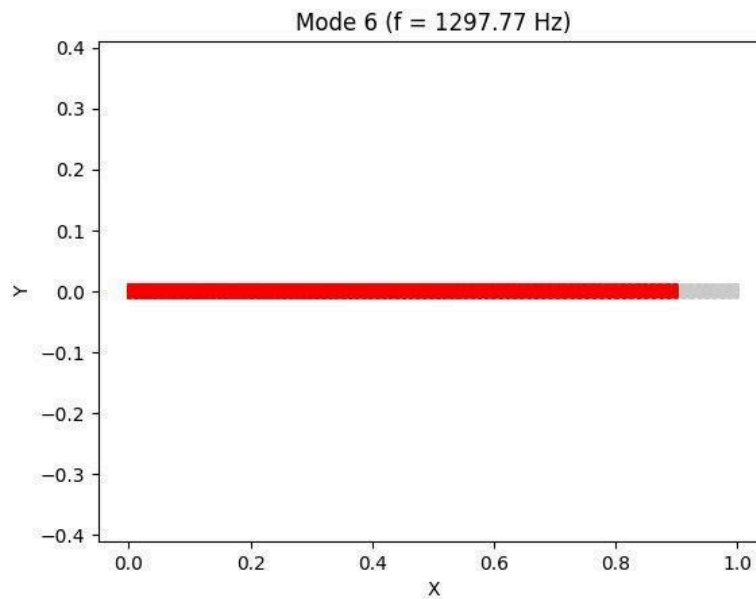


Figure 17. Mode 6: first axial (longitudinal) mode at 1297.77 Hz

Observation: Predominantly axial (horizontal) displacement with negligible transverse displacement — the beam stretches and contracts along its length. This is the first longitudinal mode, which for this geometry ( $L/H = 10$ , steel) appears between the 5th and 6th flexural modes.

## 7. Conclusion

The finite element analysis of the cantilever beam successfully captures displacement, stress, strain, and energy behavior under a uniformly distributed load. The key findings are:

- FEM results agree with Euler–Bernoulli analytical predictions to within 0.11% at 640 elements ( $AR = 10$ ).
- Mesh convergence is confirmed by both tip displacement and total strain energy, stabilising beyond ~640 elements.
- For low aspect ratios ( $AR < 5$ ), shear deformation is significant; Timoshenko beam theory is more appropriate than Euler–Bernoulli in this regime.
- Normal stress  $\sigma_x$  varies linearly across the cross-section, peaking at the fixed support, consistent with classical bending theory. Shear stress  $\tau_{xy}$  follows a near-parabolic distribution, maximum near the neutral axis and the clamped end.
- Transverse strain  $\varepsilon_y$  confirms Poisson coupling ( $\varepsilon_y = -\nu\varepsilon_x$ ), validating the plane stress assumption.
- Von Mises stress correctly identifies the clamped root as the critical failure location, consistent with maximum bending moment.
- Six natural frequencies are extracted. Modes 1–5 are flexural bending modes of increasing complexity; Mode 6 is the first longitudinal (axial) mode.
- Mode shape complexity scales with mode number: from the smooth fundamental bending curve (Mode 1) to a four-node oscillatory pattern (Mode 5).
- SolidWorks simulation results agree with the custom FEM code to within 1%, further validating the implementation.

Overall, the developed FEM code provides a reliable, accurate, and efficient tool for analyzing both the static and free-vibration characteristics of beam structures. The study confirms that appropriate mesh refinement and the correct choice of beam theory (Euler vs. Timoshenko) are critical for obtaining accurate results as a function of beam slenderness.